Deep Learning Laboratory - Assignment 3

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Model

The data contain 36 features, five classes, and train/validation/test sizes of 2000/500/500. The network is $36 \rightarrow 30 \rightarrow 15 \rightarrow 5$. For Configuration 1, $\phi = \tanh$; for Configuration 2, $\phi = \mathrm{ReLU}$.

The same simple operation is used in each hidden layer:

$$h = \phi(Wx + b).$$

The output layer uses softmax probabilities, $p = \mathrm{softmax}(z)$, and the average cross-entropy error is

$$E = -\frac{1}{N} \sum t \log p.$$

Pattern mode uses batch size 1. Mini-batch mode uses five subsets per epoch. All methods follow the basic update

$$\theta_{\mathrm{new}} = \theta_{\mathrm{old}} - \eta g,$$

where g is the gradient. The delta rule uses $\eta = 0.8$, generalized delta also keeps 0.9 of the previous update, and AdaM adjusts the step size from recent gradients. Training stops when $|E_e - E_{e-1}| < 10^{-3}$ for five consecutive epochs after epoch 20, or at the epoch limit. The best validation checkpoint is evaluated.

Baseline results

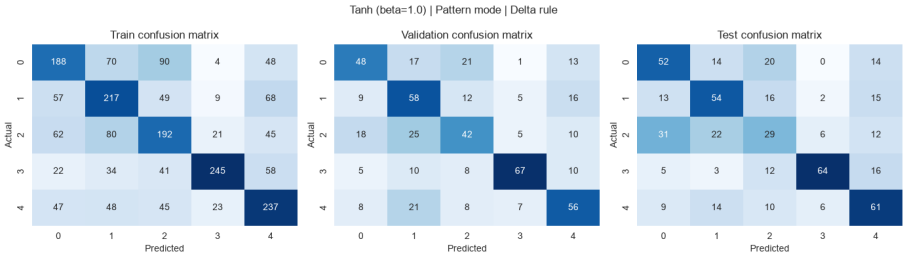
Activation	Mode / update	Epochs	Train (%)	Val. (%)	Test (%)
Tanh	Pattern / Delta	60	53.95	54.20	52.00
Tanh	Pattern / Gen. delta	60	55.70	55.80	52.40
Tanh	Pattern / AdaM	60	56.60	56.20	51.20
Tanh	Mini-batch / Delta	300	51.20	54.40	49.40
Tanh	Mini-batch / Gen. delta	300	61.10	56.20	53.20
Tanh	Mini-batch / AdaM	153	52.75	53.60	51.00
ReLU	Pattern / Delta	60	59.55	56.20	50.40
ReLU	Pattern / Gen. delta	60	59.15	**56.80**	**53.80**
ReLU	Pattern / AdaM	60	59.75	55.80	**53.80**
ReLU	Mini-batch / Delta	300	51.30	54.60	49.80
ReLU	Mini-batch / Gen. delta	300	64.20	56.40	52.80
ReLU	Mini-batch / AdaM	182	54.90	54.80	52.00

ReLU with pattern-mode generalized delta gives the highest validation accuracy. The best test accuracy, 53.8%, is shared by the two ReLU pattern-mode runs.

Error curves and confusion matrices

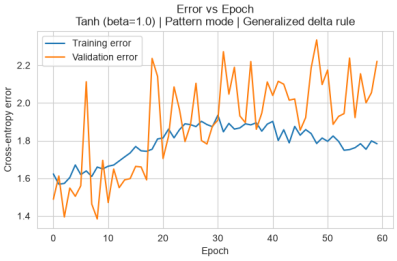
C1: Tanh - Pattern - Delta

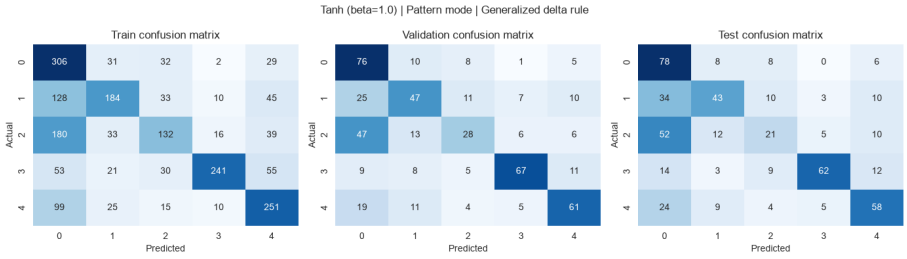




Train 53.95%, validation 54.20%, test 52.00%.

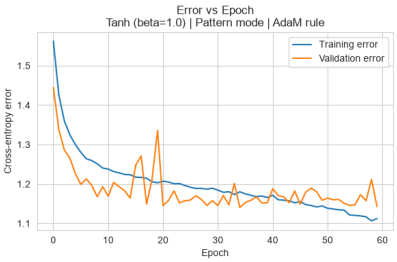
C1: Tanh - Pattern - Generalized delta

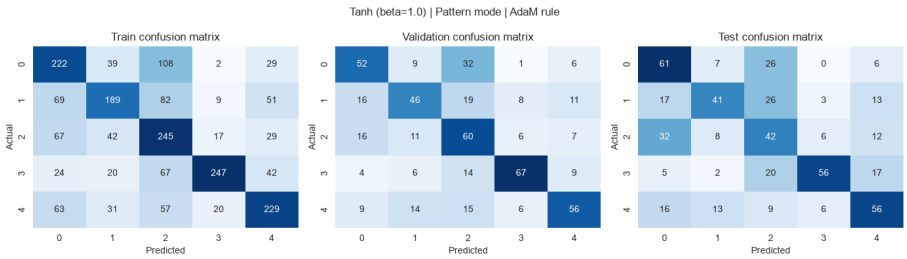




Train 55.70%, validation 55.80%, test 52.40%.

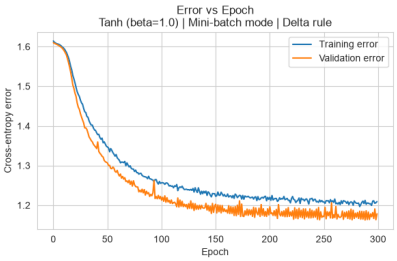
C1: Tanh - Pattern - AdaM

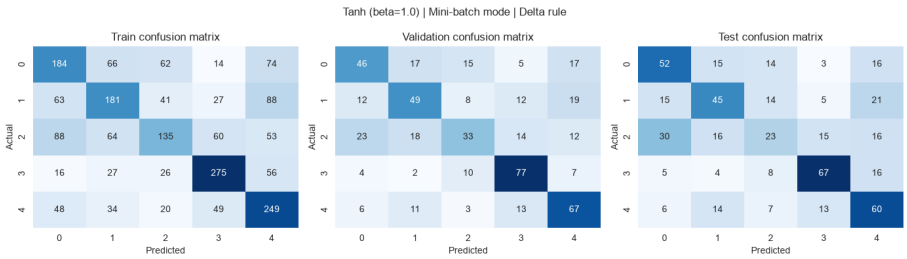




Train 56.60%, validation 56.20%, test 51.20%.

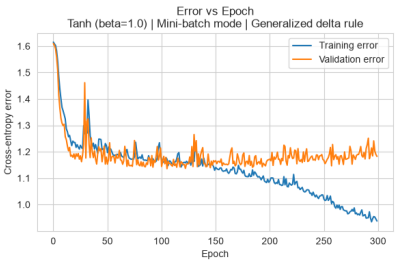
C1: Tanh - Mini-batch - Delta

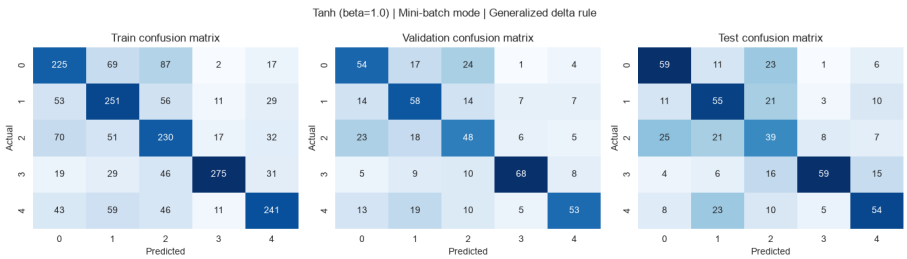




Train 51.20%, validation 54.40%, test 49.40%.

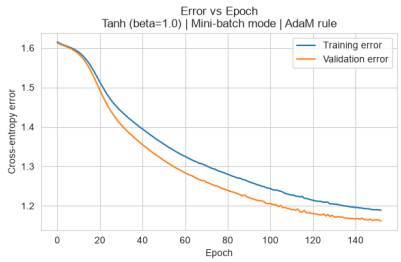
C1: Tanh - Mini-batch - Generalized delta

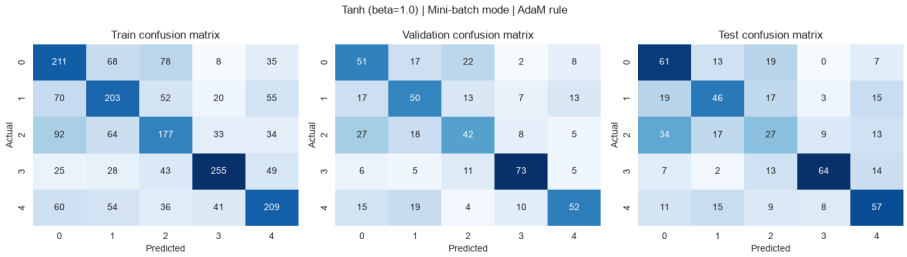




Train 61.10%, validation 56.20%, test 53.20%.

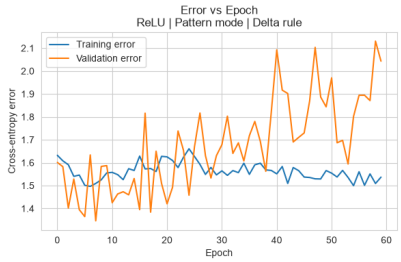
C1: Tanh - Mini-batch - AdaM

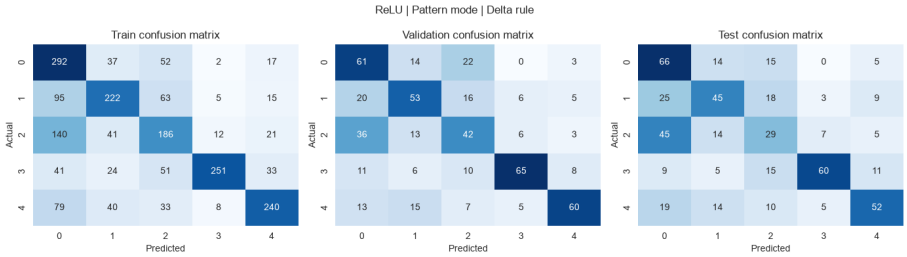




Train 52.75%, validation 53.60%, test 51.00%.

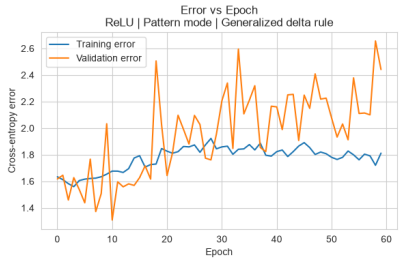
C2: ReLU - Pattern - Delta

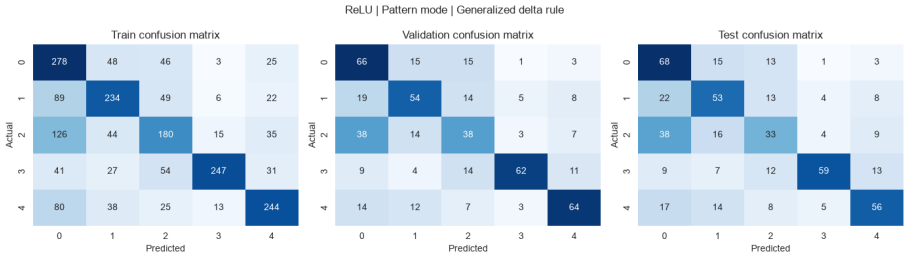




Train 59.55%, validation 56.20%, test 50.40%.

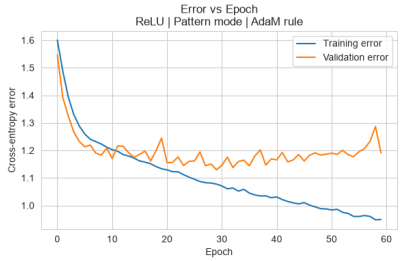
C2: ReLU - Pattern - Generalized delta

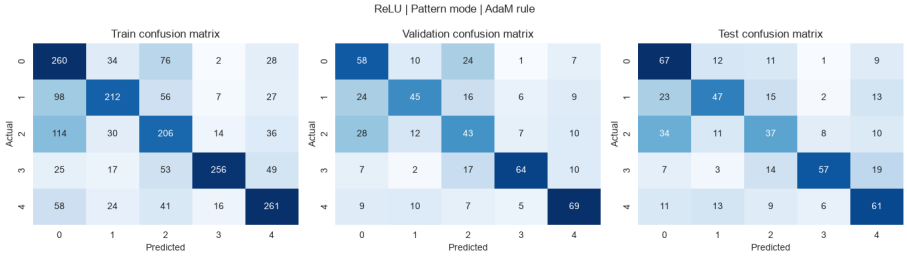




Train 59.15%, validation 56.80%, test 53.80%.

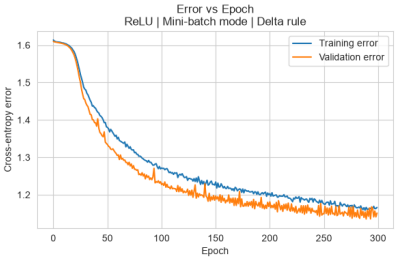
C2: ReLU - Pattern - AdaM

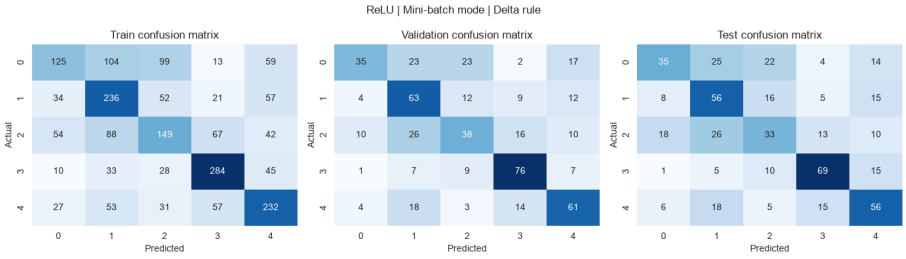




Train 59.75%, validation 55.80%, test 53.80%.

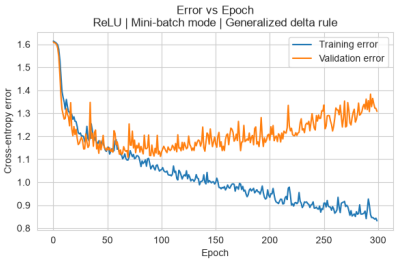
C2: ReLU - Mini-batch - Delta

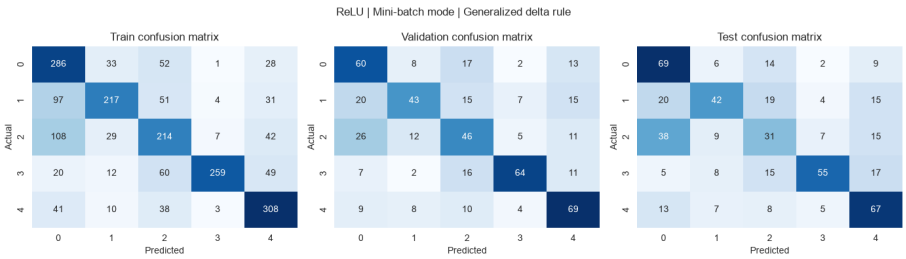




Train 51.30%, validation 54.60%, test 49.80%.

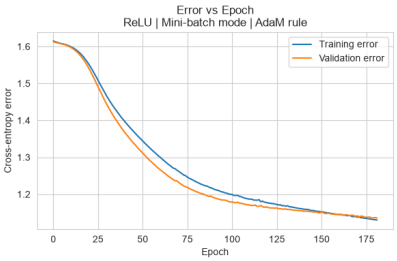
C2: ReLU - Mini-batch - Generalized delta

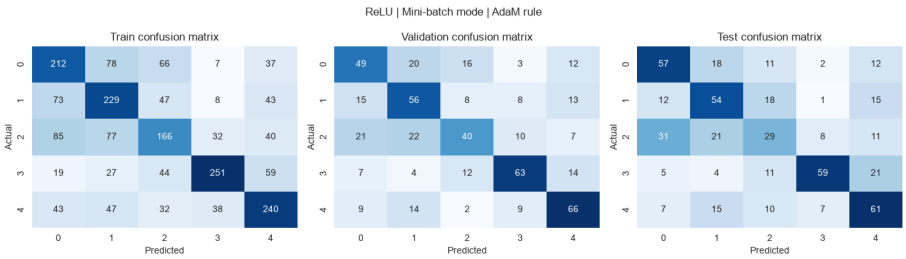




Train 64.20%, validation 56.40%, test 52.80%.

C2: ReLU - Mini-batch - AdaM





Train 54.90%, validation 54.80%, test 52.00%.

Normalization study

For mean-variance normalization, μ and σ are the training mean and standard deviation:

$$x' = \frac{x - \mu}{\sigma}.$$

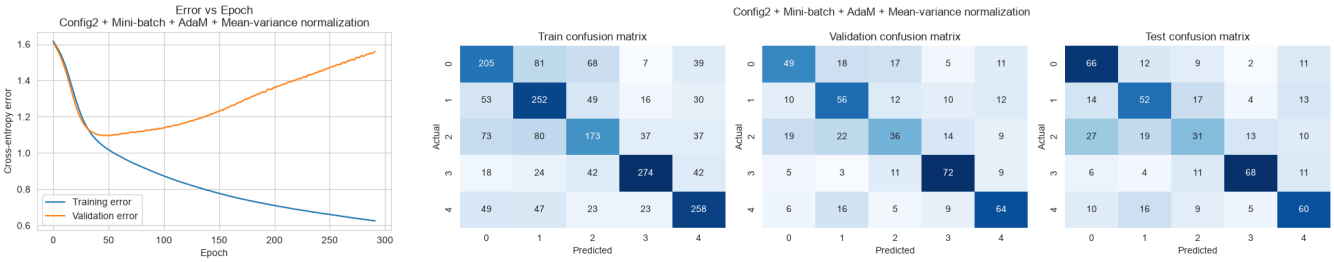
Layer normalization applies the same idea within each hidden layer:

$$a' = \frac{a - \mu_a}{\sqrt{\sigma_a^2 + \epsilon}},$$

followed by a learned scale and bias.

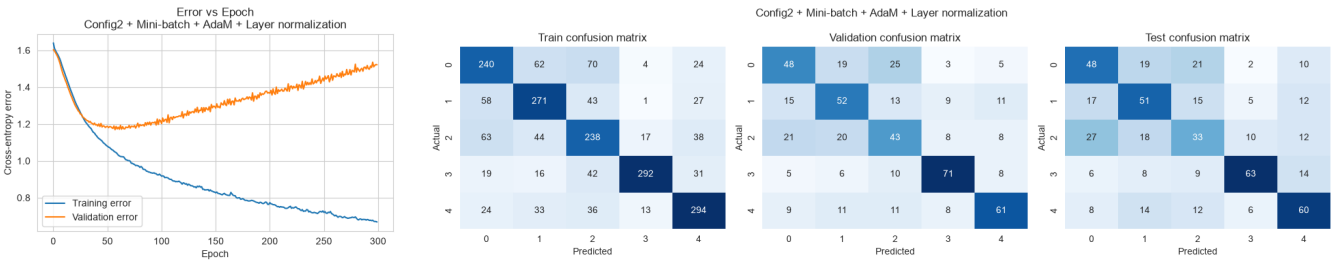
Setup	Train (%)	Val. (%)	Test (%)
No normalization	54.90	54.80	52.00
Mean-variance	58.10	**55.40**	**55.40**
Layer normalization	**66.75**	55.00	51.00

C2: Mini-batch - AdaM - Mean-variance normalization

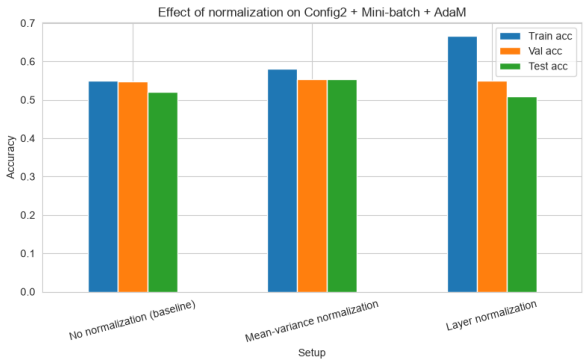


Mean-variance normalization gives the best test accuracy: 55.40%.

C2: Mini-batch - AdaM - Layer normalization



Layer normalization raises training accuracy but not test accuracy.



Conclusion. Mean-variance normalization gives the best final test result, 55.4%. Layer normalization shows the largest train-test gap.